LOAN FRAUD DETECTION PROJECT

Eulers Group

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Refactory, June 2024

INTRODUCTION

Loan Fraud is a significant concern for financial institutions, leading to substantial financial losses and reputation damage. As fraudsters become more sophisticated, there is a pressing need for a reliable and efficient system to detect and prevent fraudulent activities in loan applications.

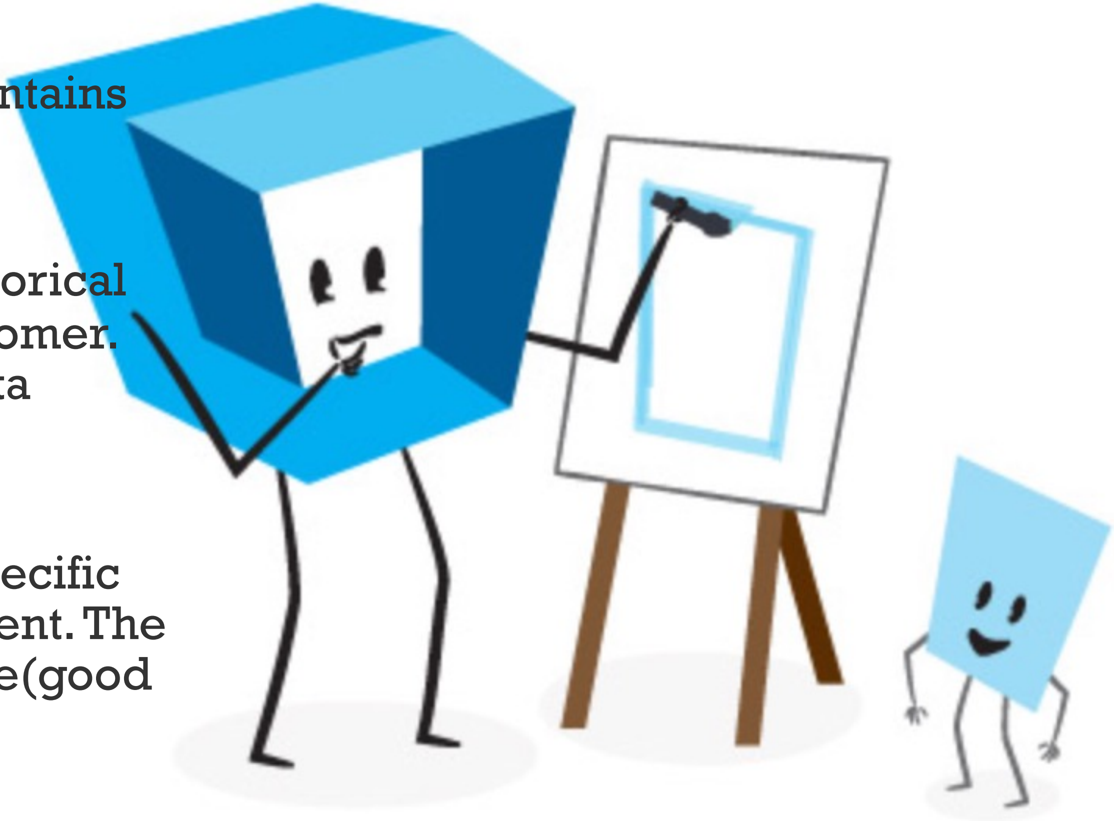
Objectives

1. To develop a predictive model that accurately determines each application's loan repayment likelihood.
2. To enhance the efficiency of loan processing by automating the detection of fraudulent activities
3. To reduce financial losses and reputational risks for financial institutions.
4. To strengthen trust between lenders and customers by ensuring fair and secure lending practices.

DATA DESCRIPTION

- The data used is approximately **14,743** rows, **28** columns and some columns contain nulls, duplicates. Test data was also provided for this challenge.

1. **Demographics Data:** (train and test demographics') Contains personal information about the borrower, such as age, employment status, location and education level.
2. **Previous Loans Data:** (train and test prev) provides historical information about loans previously accessed by the customer. The fields include loan amount, first repaid date. This data provides insight into the repayment behavior and creditworthiness.
3. **Performance Data**:(train and test perf) is the on the specific loan application from which you need to predict repayment. The data fields provided include, loan amount, target variable(good bad flag)

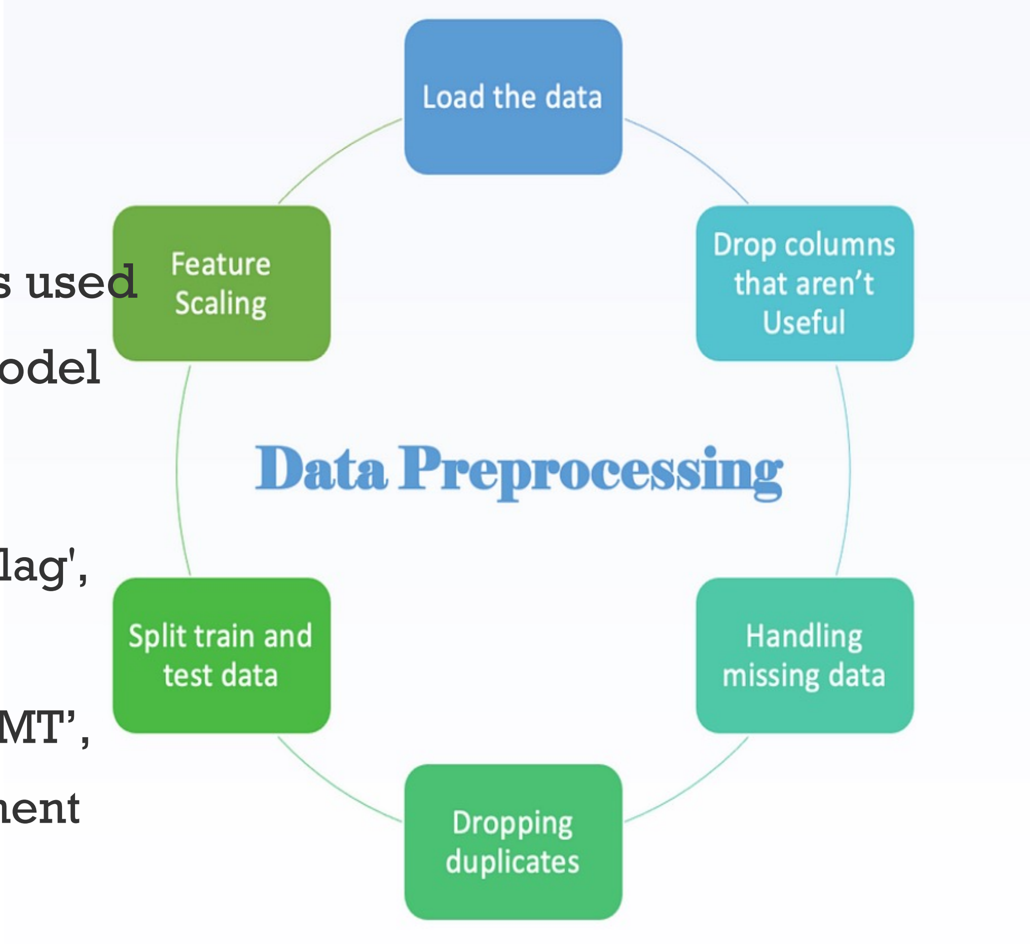


DATA PROCESSING

1. Exploratory Data Analysis (EDA) was done to understand the data and relationships. Data profiling with YData profiling

2. Duplicates were removed , nulls imputed

3. Feature engineering was done to create additional features used for prediction and a new feature was created to improve model performance in the process of hyper parameter tuning.



features derived ; 'age_category ', 'education_flag', employment_flag', 'bnk_acc_typ_flag', 'loan_freq', 'average_paybacktime', 'clients_avg_loanAMT', 'clients_max_loanAMT', 'clients_min_loanAMT', 'clients_modal_loanAMT', 'total_loans', 'ontime_loans','ontime psyment percentage', 'pays_late_risk', 'referred_client', 'loan_interest', '%loan_interest', 'loan_rateCategories','BorrowerReliabilityScore'

MODEL SELECTION AND TRAINING

1. Explored the classification algorithms (Random Forest, Light Gboost, XGBoost, Neural network) and the hyper parameters were tuned and trained on the data. XGBoost gave the the best performance score and was used to predict

1. Data was provided to us already divided into training and validation sets to prevent overfitting.

2. training set and evaluate on validation set. Calculate misclassification rate on validation, set for accuracy assessment.

MODEL EVALUATION

1. All models applied were evaluated using accuracy, F1, recall, precision, recall and AUC. The performance of the various models while tuning the hyperparameters was compared and the model with the best accuracy score of 0.77 was adopted and submitted, resulting in a 0.215 score and 123rd position on the Leaderboard.
2. Feature importance was determined

```
accuracy score:  0.7368421052631579
f1 score:  0.8389355742296917
precision score:  0.7881578947368421
recall score:  0.8967065868263473
auc:  0.6122064124178827
```

```
Perf result Xtrain using smote data - poor pe
accuracy score:  0.7093821510297483
f1 score:  0.812407680945347
precision score:  0.8017492711370262
recall score:  0.8233532934131736
auc:  0.621798877972211
```

```
Perf result Xtrain excludes features of 0.1 a
accuracy score:  0.7356979405034325
f1 score:  0.8381219341275403
precision score:  0.7878787878787878
recall score:  0.8952095808383234
auc:  0.6135907796058369
```

```
Perf result Xtrain.........
accuracy score:  0.7700228832951945
f1 score:  0.8659106070713809
precision score:  0.7809867629362214
recall score:  0.9715568862275449
auc:  0.6135907796058369
```

Competition Leaderboard

Unless stated otherwise in the Info Page, this leaderboard reflects scores based on only a portion of the total test set until the competition closes. See competition Info for more information.

RANK	USER	PUBLIC SCORE
123	Eulers Team	0.215862068


Thank you